



Continuous-Discrete preconditioned Crank Nicholson

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May 6, 2026

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Chapter 1

Introduction

1.1 Feynman–Kac Models

Feynman–Kac models provide a convenient framework for describing filtering and smoothing distributions in latent dynamical systems. Let $(\mathcal{X}_t, X_t)$, $t = 1, \dots, T$, be measurable state spaces. A Feynman–Kac model is specified by an initial law $M_1(dx_1)$, Markov transition kernels $M_t(x_{t-1}, dx_t)$ for $t \geq 2$, and potential functions $G_1(x_1)$ and $G_t(x_{t-1}, x_t)$ for $t \geq 2$. These define the path-space law

$$\pi_t(dx_{1:t}) \propto G_1(x_1)M_1(dx_1) \prod_{s=2}^t G_s(x_{s-1}, x_s)M_s(x_{s-1}, dx_s), \quad (1.1)$$

The corresponding filtering distribution is the marginal of $\pi_t(dx_{1:t})$ at x_t , which we denote by $\eta_t(dx_t)$. The principal Feynman–Kac model is the state-space model. Let $\mathcal{X} = \mathbb{R}^{d_x}$ and $\mathcal{Y} = \mathbb{R}^{d_y}$, then the latent process and observations up to time t satisfy the joint posterior distribution

$$p(x_{1:t} \mid y_{1:t}) \propto f_1(x_1)h_1(y_1 \mid x_1) \prod_{s=2}^t f_s(x_s \mid x_{s-1})h_s(y_s \mid x_s). \quad (1.2)$$

This can be written in Feynman–Kac form in several ways [5]. The simplest choice is the bootstrap formulation,

$$M_1(dx_1) = f_1(x_1) dx_1, \quad (1.3)$$

$$M_t(x_{t-1}, dx_t) = f_t(x_t \mid x_{t-1}) dx_t, \quad t \geq 2, \quad (1.4)$$

$$G_1(x_1) = h_1(y_1 \mid x_1), \quad (1.5)$$

$$G_t(x_{t-1}, x_t) = h_t(y_t \mid x_t), \quad t \geq 2. \quad (1.6)$$

In this case, the mutation kernel coincides with the latent dynamics, and the potentials depend only on the current state. More generally, one may choose any evaluable proposal kernel M_t satisfying absolute continuity with respect to the latent dynamics, and define the potential so

that the product $M_t G_t$ still recovers the desired target. In density notation, this gives the guided formulation

$$G_t(x_{t-1}, x_t) = \frac{f_t(x_t \mid x_{t-1}) h_t(y_t \mid x_t)}{M_t(x_t \mid x_{t-1})}, \quad (1.7)$$

with the analogous expression at $t = 1$. This is useful because it allows the proposal to incorporate information from the observations while preserving the same target distribution. In practice, well-chosen guided proposals can reduce weight degeneracy and improve the efficiency of particle-based inference [5].

1.2 Sequential Monte Carlo

Sequential Monte Carlo (SMC) methods approximate the Feynman–Kac recursion by a weighted particle system. Given particles $\{X_t^i\}_{i=1}^N$ and normalised weights $\{W_t^i\}_{i=1}^N$, expectations under the filtering law may be approximated by

$$\eta_t^N(\psi) := \sum_{i=1}^N W_t^i \psi(X_t^i). \quad (1.8)$$

Equivalently, the filtering distribution is represented by the weighted empirical measure

$$\eta_t^N(dx_t) := \sum_{i=1}^N W_t^i \delta_{X_t^i}(dx_t). \quad (1.9)$$

The particle filter proceeds forward in time. At each step, particles are propagated using the mutation kernel, weighted by the potential, and resampled in order to concentrate computational effort on regions of high posterior probability. This yields an online approximation of the filtering distributions $\eta_t(dx_t)$. The corresponding forward pass is given in Algorithm 1.

For brevity, particle-wise loops are suppressed in Algorithms 1–6; any operation indexed by i is to be understood as being performed for all particles in $\{1, \dots, N\}$.

For offline inference, the quantity of interest is often the smoothing distribution over the full latent path. In contrast to filtering, smoothing uses the entire dataset $y_{1:T}$ to infer earlier latent states. A naive way to obtain a smoothing trajectory from particle filter output is to sample a terminal particle and trace its ancestors backwards through the stored genealogy. However, repeated resampling causes severe path degeneracy: at earlier times many particles share the same descendant, so the resulting path sample may collapse onto very few distinct trajectories. A more effective alternative is backward sampling, which reweights particles at time $t-1$ according to how plausibly they could have generated the selected future state and thereby exploits additional smoothing information. The corresponding backward pass is given in Algorithm 2 [5].

Algorithm 1 Particle Filter (Forward Pass)

Input: Number of particles N , $M_1(\mathrm{d}x_1)$, $\{M_t(x_{t-1}, \mathrm{d}x_t)\}_{t=2}^T$, $\{G_t\}_{t=1}^T$

Output: Particles $\{X_t^{1:N}\}_{t=1}^T$, ancestors $\{A_t^{1:N}\}_{t=2}^T$, weights $\{W_t^{1:N}\}_{t=1}^T$

- 1: Sample $X_1^i \sim M_1(\mathrm{d}x_1)$
 - 2: Set $w_1^i = G_1(X_1^i)$
 - 3: Normalise weights: $W_1^i = w_1^i / \sum_{j=1}^N w_1^j$
 - 4: **for** $t = 2, \dots, T$ **do**
 - 5: Sample $A_t^i \sim \mathcal{M}(W_{t-1}^{1:N})$
 - 6: Sample $X_t^i \sim M_t(X_{t-1}^{A_t^i}, \mathrm{d}x_t)$
 - 7: Set $w_t^i \leftarrow G_t(X_{t-1}^{A_t^i}, X_t^i)$
 - 8: Normalise weights: $W_t^i = w_t^i / \sum_{j=1}^N w_t^j$
 - 9: **end for**
-

Algorithm 2 Particle Smoother (Backward Sampling Pass)

Input: Particles $\{X_t^{1:N}\}_{t=1}^T$, weights $\{W_t^{1:N}\}_{t=1}^T$, $\{M_t(x_{t-1}, \mathrm{d}x_t)\}_{t=2}^T$, $\{G_t\}_{t=2}^T$

Output: Indices $B_{1:T}$ and path $\{X_t^{B_t}\}_{t=1}^T$ approximately from the smoothing distribution

- 1: Sample $B_T \sim \mathcal{M}(W_T^{1:N})$
 - 2: **for** $t = T, \dots, 2$ **do**
 - 3: Set $\tilde{w}_{t-1}^i = W_{t-1}^i \cdot G_t(X_{t-1}^i, X_t^{B_t}) \cdot M_t(X_{t-1}^i, X_t^{B_t})$
 - 4: Normalise weights: $\tilde{W}_{t-1}^i = \tilde{w}_{t-1}^i / \sum_{j=1}^N \tilde{w}_{t-1}^j$
 - 5: Sample $B_{t-1} \sim \mathcal{M}(\tilde{W}_{t-1}^{1:N})$
 - 6: **end for**
-

1.2.1 Conditional SMC

Algorithms 1 and 2 can be used to produce a single particle approximation to the smoothing distribution. However, when the number of particles is limited, or when the latent process is high-dimensional, a single particle smoothing pass may provide a poor approximation to the full smoothing law. Particle MCMC methods address this by constructing Markov kernels whose invariant distribution is the target smoothing distribution [1].

A key ingredient is *conditional sequential Monte Carlo* (CSMC). In CSMC, one conditions the particle system on a retained reference trajectory

$$\bar{x}_{1:T} = (\bar{x}_1, \dots, \bar{x}_T), \quad (1.10)$$

typically the current state of the outer Markov chain. The particle filter is then modified so that this trajectory is preserved throughout the forward pass. Concretely, one particle at each time is forced to coincide with the corresponding component of the reference path, while the remaining particles are sampled in the usual way. This ensures that the previous trajectory always remains in the support of the update, which is essential for constructing a valid particle MCMC kernel.

When combined with a terminal selection or a backward-sampling pass, CSMC defines a Markov kernel on trajectories that leaves the smoothing distribution invariant. In particular, using backward sampling substantially improves mixing relative to simple genealogy tracing, since it allows

the new trajectory to depart more freely from the retained reference path. The resulting CSMC kernel is given in Algorithm 3. As above, particle-wise operations indexed by i are understood to be carried out for all particles in $\{1, \dots, N\}$. Since the ordering of particles is immaterial, we may, without loss of generality, place the retained reference particle at index 1 at every time step.

Algorithm 3 CSMC

Input: Reference path $\bar{x}_{1:T}$, number of particles N , $M_1(dx_1)$, $\{M_t(x_{t-1}, dx_t)\}_{t=2}^T$, $\{G_t\}_{t=1}^T$

Output: New trajectory $\tilde{x}_{1:T}$ obtained from a π_T -invariant Markov kernel

```

1: Set  $X_1^1 = \bar{x}_1$ 
2: Sample  $X_1^i \sim M_1(dx_1)$   $i \neq 1$ 
3: Set  $w_1^i = G_1(X_1^i)$ 
4: Normalise weights:  $W_1^i = w_1^i / \sum_{j=1}^N w_1^j$ 
5: for  $t = 2, \dots, T$  do
6:   Set  $A_t^1 = 1$ 
7:   Set  $X_t^1 = \bar{x}_t$ 
8:   Sample  $A_t^i \sim \mathcal{M}(W_{t-1}^{1:N})$   $i \neq 1$ 
9:   Sample  $X_t^i \sim M_t(X_{t-1}^{A_t^i}, dx_t)$   $i \neq 1$ 
10:  Set  $w_t^i = G_t(X_{t-1}^{A_t^i}, X_t^i)$ 
11:  Normalise weights:  $W_t^i = w_t^i / \sum_{j=1}^N w_t^j$ 
12: end for
13: Sample  $B_T \sim \mathcal{M}(W_T^{1:N})$ 
14: for  $t = T, \dots, 2$  do
15:   Set  $\tilde{w}_{t-1}^i = W_{t-1}^i \cdot G_t(X_{t-1}^i, X_t^{B_t}) \cdot M_t(X_{t-1}^i, X_t^{B_t})$ 
16:   Normalise weights:  $\tilde{W}_{t-1}^i = \tilde{w}_{t-1}^i / \sum_{j=1}^N \tilde{w}_{t-1}^j$ 
17:   Sample  $B_{t-1} \sim \mathcal{M}(\tilde{W}_{t-1}^{1:N})$ 
18: end for
19: Return  $\tilde{x}_{1:T} = (X_1^{B_1}, \dots, X_T^{B_T})$ 

```

1.3 Local Particle Methods

Although CSMC scales favourably with the time horizon T relative to standard path-space MCMC methods, it still suffers from a curse of dimensionality in the latent-state dimension D . In high dimensions, the conditional particle system is increasingly likely to reproduce the reference trajectory, so that the resulting CSMC Markov kernel exhibits poor mixing. Intuitively, this occurs because the particles are proposed independently of the current state component $\bar{x}_t$, and such global proposals become increasingly ineffective as D grows [4, 2].

A natural remedy is to replace global proposals with *local* proposals centred around the current reference path. Recent work has shown that this can be achieved by introducing auxiliary variables, interpreted as pseudo-observations, and then running a standard CSMC update on an augmented

Feynman–Kac model [4, 3, 2]. Herein, we define the following:

$$Q_1(x_1) = G_1(x_1)M_1(dx_1) \quad (1.11)$$

$$Q_t(x_{t-1:t}) = G_t(x_{t-1}, x_t)M_t(dx_t|x_{t-1}) \quad t \geq 2 \quad (1.12)$$

such that the original Feynman–Kac target is written (with some abuse of notation)

$$\pi_T(x_{1:T}) \propto \prod_{t=1}^T Q_t(x_{t-1:t}), \quad (1.13)$$

We now describe the simplest local particle construction: the particle random-walk Metropolis (Particle-RWM) algorithm.

1.3.1 Particle Random-Walk Metropolis

Particle-RWM was introduced by [4] and later interpreted through an auxiliary-variable formulation by [3]. The idea is to replace the global mutation step of CSMC by a local Gaussian perturbation around the current reference state x_t .

Algorithm 4 Particle-RWM. CSMC but with steps 2, 3, 9, and 10 modified as follows

- 2: Sample $\phi_1 \mid \bar{x}_1 \sim \mathcal{N}(\bar{x}_1, \frac{\delta_1}{2}I)$ and $x_1^i \mid \phi_1 \sim \mathcal{N}(\phi_1, \frac{\delta_1}{2}I)$, $i \neq 1$
 - 3: Set $w_1^i = Q_1(X_1^i)$
 - 9: Sample $\phi_t \mid \bar{x}_t \sim \mathcal{N}(\bar{x}_t, \frac{\delta_t}{2}I)$ and $x_t^i \mid \phi_t \sim \mathcal{N}(\phi_t, \frac{\delta_t}{2}I)$, $i \neq 1$
 - 10: Set $w_t^i = Q_t(X_{t-1}^{A^i}, X_t^i)$
-

Marginally, this yields $x_t^i \mid x_t \sim \mathcal{N}(x_t, \delta_t I)$ when $i \neq 1$ which shows clearly that the proposed particles are local perturbations of the reference path. The scale parameter $\delta_t > 0$ therefore controls the locality of the proposal, and hence the resulting acceptance behaviour of the induced MCMC kernel [2].

The auxiliary-variable construction is based on the augmented target

$$\pi'_T(x_{1:T}, \phi_{1:T}) := \pi_T(x_{1:T}) \prod_{t=1}^T \mathcal{N}\left(\phi_t; x_t, \frac{\delta_t}{2}I\right) \quad (1.14)$$

$$\propto \prod_{t=1}^T Q_t(x_{t-1:t}) \mathcal{N}\left(\phi_t; x_t, \frac{\delta_t}{2}I\right). \quad (1.15)$$

which clearly admits $\pi_T(x_{1:T})$ as its marginal in $x_{1:T}$. To cast this as a new Feynman–Kac model, define

$$G'_t(x_{t-1:t}) := Q_t(x_{t-1:t}) \quad (1.16)$$

$$M'_t(x_t \mid x_{t-1}; \phi_t) := \mathcal{N}\left(x_t; \phi_t, \frac{\delta_t}{2}I\right). \quad (1.17)$$

Since Gaussian densities are symmetric in their arguments, $\mathcal{N}(x_t; \phi_t, \frac{\delta_t}{2}I) = \mathcal{N}(\phi_t; x_t, \frac{\delta_t}{2}I)$, we obtain

$$Q'_t(x_{t-1:t}; \phi_t) = G'(x_{t-1:t})M'_t(x_t | x_{t-1}) \quad (1.18)$$

$$:= Q_t(x_{t-1:t})\mathcal{N}\left(\phi_t; x_t, \frac{\delta_t}{2}I\right). \quad (1.19)$$

This form is important conceptually. The auxiliary variable ϕ_t is introduced only to construct a local proposal around the current reference state. Once we condition on ϕ_t , the Gaussian pseudo-observation factor is already represented inside the proposal kernel M'_t . Consequently, no additional correction involving ϕ_t is needed in the potential; the potential remains exactly the original incremental target factor Q_t . In this sense, Particle-RWM is simply a standard CSMC update applied to a modified Feynman–Kac model on an extended space [3, 2].

In particular, the Particle-RWM algorithm can be interpreted as alternating two steps:

$$\phi_t | x_t \sim \mathcal{N}\left(x_t, \frac{\delta_t}{2}I\right), \quad t = 1, \dots, T, \quad (1.20)$$

$$x'_{1:T} | \phi_{1:T} \sim \text{CSMC on the Feynman–Kac model } \{Q'_t(\cdot; \phi_t)\}_{t=1}^T. \quad (1.21)$$

Since the augmented target has marginal $\pi_T(x_{1:T})$, this defines a valid π_T -invariant Markov kernel [1, 6, 3, 2].

1.4 Continuous-Discrete State-Space Models

We now turn to continuous-discrete state-space models (CD-SSMs), in which the latent process evolves in continuous time but is observed only at discrete times. Let $0 = s_0 < s_1 < \dots < s_T$, and let the latent signal $X = \{X(s)\}_{s \geq 0}$ solve the stochastic differential equation

$$dX(s) = b(s, X(s)) ds + \sigma(s, X(s)) dB(s), \quad X(0) = x(0) \in \mathbb{R}^d. \quad (1.22)$$

At observation times s_t , noisy measurements Y_t are generated according to

$$Y_t | X(s_t) \sim f_t(\cdot | X(s_t)), \quad t = 1, \dots, T. \quad (1.23)$$

A natural way to place this model into the Feynman–Kac framework is to regard the latent state at step t as the continuous path segment over the interval $[s_{t-1}, s_t]$, together with its end-point. Writing

$$V_t(s) = X(s_{t-1} + s), \quad s \in [0, \Delta_t], \quad (1.24)$$

$$E_t = V_t(\Delta_t), \quad (1.25)$$

with $\Delta_t = s_t - s_{t-1}$, the latent variable may be represented as $X_t = (V_t, E_t)$. The corresponding posterior can then be written as a discrete-time state-space model on path space [5].

However, the continuous-time nature of the latent process creates a major obstacle for smoothing. In standard backward sampling, one attempts to reselect the ancestor of a future particle at time t from among all particles at time $t-1$. In a CD-SSM this is typically impossible: a continuous path over $[s_{t-1}, s_t]$ must start exactly where the previous segment ends, so changing the ancestor would create a discontinuity that has zero probability under the model. Consequently, the only admissible ancestor is the one already selected in the forward pass, and backward sampling collapses to trivial genealogy tracing [5].

Stanton and Beskos overcome this problem by reparameterising each path segment. Rather than working directly with the path V_t , they introduce transformed variables under which both target and proposal admit densities with respect to a common reference measure of the form $\text{Leb}(\text{d}e_t) \otimes \mathbb{W}_t(\text{d}u_t)$, where $\mathbb{W}_t$ is a Wiener measure on the driving Brownian path. This is exactly the structural condition required by FFBS and related ancestor-resampling methods [5].

Herein, we focus on the backward transform, which remains applicable even in the hypo-elliptic setting. Under this transform, the latent variable at time t becomes $z_t = (u_t, e_t)$, where $e_t = X(s_t)$ is the end-point of the latent bridge over $[s_{t-1}, s_t]$, and u_t is the Wiener path used to reconstruct that bridge through a measurable map

$$v_t = H_t(u_t; e_{t-1}, e_t). \quad (1.26)$$

Under the backward transform of Stanton and Beskos, the proposal kernel factorises as

$$\bar{M}_t^\leftarrow(z_{t-1}, \text{d}z_t) = (m_t^\leftarrow(e_t \mid e_{t-1}) \text{d}e_t) \otimes \mathbb{W}_t(\text{d}u_t), \quad (1.27)$$

so the Wiener component is sampled from a fixed reference law $\mathbb{W}_t$, independently of the ancestor, while the end-point is proposed from a tractable density $m_t^\leftarrow(e_t \mid e_{t-1})$. The transformed CD-SSM therefore admits the Feynman–Kac representation

$$\pi(z_{1:T} \mid y_{1:T}) \propto \bar{M}_1^\leftarrow(\text{d}z_1) \bar{G}_1^\leftarrow(z_1) \prod_{t=2}^T \bar{M}_t^\leftarrow(z_{t-1}, \text{d}z_t) \bar{G}_t^\leftarrow(z_{t-1}, z_t), \quad (1.28)$$

where the potential is

$$\bar{G}_t^\leftarrow(z_{t-1}, z_t) = \frac{p_t(e_t \mid e_{t-1})}{m_t^\leftarrow(e_t \mid e_{t-1})} \frac{\text{d}P_t[(e_{t-1}, e_t), \cdot]}{\text{d}M_t^\leftarrow[(e_{t-1}, e_t), \cdot]} (H_t(u_t; e_{t-1}, e_t)) f_t(y_t \mid e_t). \quad (1.29)$$

For $t = 1$, $e_0 = X(0)$ is some fixed starting point. Here $P_t[(e_{t-1}, e_t), \cdot]$ denotes the true bridge law and $M_t^\leftarrow[(e_{t-1}, e_t), \cdot]$ the chosen backward bridge proposal. In our implementation, $m_t^\leftarrow(e_t \mid e_{t-1})$ is taken to be the end-point proposal induced by a single Euler step, though the formal development does not depend on this specific choice.

Algorithm 5 Backward-Transformed Particle Filter (Forward Pass)

Input: N , $e_0 = x(0)$, $\{m_t^\leftarrow(e_t \mid e_{t-1})\}_{t=1}^T$, bridge maps $\{H_t(\cdot; e_{t-1}, e_t)\}_{t=1}^T$, $\{\bar{G}_t^\leftarrow\}_{t=1}^T$

Output: Particles $\{Z_t^{1:N}\}_{t=1}^T$, paths $\{V_t^{1:N}\}_{t=1}^T$, ancestors $\{A_t^{1:N}\}_{t=2}^T$, weights $\{W_t^{1:N}\}_{t=1}^T$

1: Sample $E_1^i \sim m_1^\leftarrow(\text{de}_1 \mid e_0)$

2: Sample $U_1^i \sim W_1(\text{du}_1)$

3: Set $Z_1^i = (U_1^i, E_1^i)$

4: Reconstruct $V_1^i = H_1(U_1^i; e_0, E_1^i)$

5: Set

$$w_1^i = \bar{G}_1^\leftarrow(Z_1^i) = \frac{p_1(E_1^i \mid e_0)}{m_1^\leftarrow(E_1^i \mid e_0)} \frac{dP_1[(e_0, E_1^i), \cdot]}{dM_1^\leftarrow[(e_0, E_1^i), \cdot]}(V_1^i) f_1(y_1 \mid E_1^i)$$

6: Normalise weights: $W_1^i = w_1^i / \sum_{j=1}^N w_1^j$

7: **for** $t = 2, \dots, T$ **do**

8: Sample $A_t^i \sim \mathcal{M}(W_{t-1}^{1:N})$

9: Sample $E_t^i \sim m_t^\leftarrow(\text{de}_t \mid E_{t-1}^{A_t^i})$

10: Sample $U_t^i \sim W_t(\text{du}_t)$

11: Set $Z_t^i = (U_t^i, E_t^i)$

12: Reconstruct $V_t^i = H_t(U_t^i; E_{t-1}^{A_t^i}, E_t^i)$

13: Set

$$w_t^i = \bar{G}_t^\leftarrow(Z_{t-1}^{A_t^i}, Z_t^i) = \frac{p_t(E_t^i \mid E_{t-1}^{A_t^i})}{m_t^\leftarrow(E_t^i \mid E_{t-1}^{A_t^i})} \frac{dP_t[(E_{t-1}^{A_t^i}, E_t^i), \cdot]}{dM_t^\leftarrow[(E_{t-1}^{A_t^i}, E_t^i), \cdot]}(V_t^i) f_t(y_t \mid E_t^i)$$

14: Normalise weights: $W_t^i = w_t^i / \sum_{j=1}^N w_t^j$

15: **end for**

Algorithm 6 Backward-Transformed Particle Smoother (Backward Sampling Pass)

Input: Particles $\{Z_t^{1:N}\}_{t=1}^T$, paths $\{V_t^{1:N}\}_{t=1}^T$, weights $\{W_t^{1:N}\}_{t=1}^T$, $\{m_t^\leftarrow(e_t \mid e_{t-1})\}_{t=2}^T$, $\{\bar{G}_t^\leftarrow\}_{t=2}^T$

Output: Indices $B_{1:T}$, trajectory $\{(V_t^{B_t}, E_t^{B_t})\}_{t=1}^T$ approximately from the smoothing distribution

1: Sample $B_T \sim \mathcal{M}(W_T^{1:N})$

2: **for** $t = T, \dots, 2$ **do**

3: Set

$$\begin{aligned} \tilde{w}_{t-1}^i &= W_{t-1}^i \cdot \bar{M}_t^\leftarrow(Z_t^{B_t} \mid Z_{t-1}^i) \cdot \bar{G}_t^\leftarrow(Z_{t-1}^i, Z_t^{B_t}) \\ &= W_{t-1}^i \cdot m_t^\leftarrow(E_t^{B_t} \mid E_{t-1}^i) \cdot \bar{G}_t^\leftarrow(Z_{t-1}^i, Z_t^{B_t}) \end{aligned}$$

4: Normalise weights: $\tilde{W}_{t-1}^i = \tilde{w}_{t-1}^i / \sum_{j=1}^N \tilde{w}_{t-1}^j$

5: Sample $B_{t-1} \sim \mathcal{M}(\tilde{W}_{t-1}^{1:N})$

6: **end for**

Chapter 2

CD-pCN

2.1 Pre-conditioned Crank Nicholson (pCN) for CD-SSMs

We now develop a local particle method for the class of backward-transformed CD-SSMs introduced above. At each observation time, the transformed latent variable is

$$z_t = (u_t, e_t),$$

where u_t denotes the Wiener-path component and e_t the end-point. Under the backward-transformed formulation, inference proceeds through T successive importance-sampling steps on these paired variables. The reference proposal is given by sampling u_t from the Wiener law $\mathbb{W}_t$ and e_t from the tractable end-point proposal $m_t^{\leftarrow}(\mathrm{d}e_t \mid e_{t-1})$.

For each $t = 1, \dots, T$, let $\mathbb{P}_t$ denote the pCN transition kernel on the Wiener space with invariant measure $\mathbb{W}_t$; that is,

$$\phi_t^u \mid u_t \sim \mathbb{P}_t(\cdot \mid u_t) \tag{2.1}$$

$$\text{s.t. } \phi_t^u = \rho_t u_t + \sqrt{1 - \rho_t^2} \xi_t, \quad \xi_t \sim \mathbb{W}_t, \tag{2.2}$$

with $\rho_t \in (0, 1)$. Given a retained reference trajectory

$$\bar{z}_{1:T} = \{(\bar{u}_t, \bar{e}_t)\}_{t=1}^T, \tag{2.3}$$

the local construction is obtained by introducing auxiliary variables and then applying CSMC to the induced conditional Feynman–Kac model. For the Wiener component, we use a pCN move centred on the retained reference path; for the end-point component, we use the same Gaussian random-walk construction as in Particle-RWM. This yields a local particle kernel on the transformed state space.

Introduce auxiliary variables

$$\phi_t = (\phi_t^u, \phi_t^e), \quad t = 1, \dots, T. \quad (2.4)$$

Recall that the backward-transformed Feynman–Kac model can be written through the unnormalised transition

$$Q_t^\leftarrow(z_{t-1:t}) = \bar{M}_t^\leftarrow(z_{t-1}, dz_t) \bar{G}_t^\leftarrow(z_{t-1}, z_t), \quad (2.5)$$

where $\bar{M}_t^\leftarrow$ is the backward proposal kernel and $\bar{G}_t^\leftarrow$ is the corresponding potential. We then define the augmented target by

$$\pi'_T(z_{1:T}, \phi_{1:T}) := \pi_T(z_{1:T}) \prod_{t=1}^T \mathbb{P}_t(d\phi_t^u | u_t) \mathcal{N}\left(\phi_t^e; e_t, \frac{\delta_t}{2} I\right) \quad (2.6)$$

$$\propto \prod_{t=1}^T Q_t^\leftarrow(z_{t-1:t}) \mathbb{P}_t(d\phi_t^u | u_t) \mathcal{N}\left(\phi_t^e; e_t, \frac{\delta_t}{2} I\right) \quad (2.7)$$

$$= \prod_{t=1}^T m_t^\leftarrow(e_t | e_{t-1}) de_t \bar{G}_t^\leftarrow(z_{t-1}, z_t) [\mathbb{P}_t(d\phi_t^u | u_t) \mathbb{W}_t(du_t)] \mathcal{N}\left(\phi_t^e; e_t, \frac{\delta_t}{2} I\right), \quad (2.8)$$

where the final line uses the factorisation of the backward proposal in Equation (1.27). By construction, the marginal of π'_T in the variables $z_{1:T}$ is exactly π_T .

Algorithm 7 CD-pCN

Input: Reference trajectory $\bar{z}_{1:T} = \{(\bar{u}_t, \bar{e}_t)\}_{t=1}^T$, number of particles N , step sizes $\{\delta_t\}_{t=1}^T$, pCN kernels $\{\mathbb{P}_t\}_{t=1}^T$, and backward incremental weights $\{Q_t^\leftarrow\}_{t=1}^T$

Output: New trajectory $\tilde{z}_{1:T}$ obtained from a π_T -invariant Markov kernel

1: **for** $t = 1, \dots, T$ **do**

2: Sample

$$\phi_t^u | \bar{u}_t \sim \mathbb{P}_t(\cdot | \bar{u}_t), \quad \phi_t^e | \bar{e}_t \sim \mathcal{N}\left(\bar{e}_t, \frac{\delta_t}{2} I\right).$$

3: **end for**

4: Define, for $t = 1, \dots, T$,

$$M'_t(z_{t-1}, dz_t; \phi_t) := \mathbb{P}_t(du_t | \phi_t^u) \mathcal{N}\left(e_t; \phi_t^e, \frac{\delta_t}{2} I\right) de_t,$$

$$G'_t(z_{t-1}, z_t; \phi_t) := Q_t^\leftarrow(z_{t-1:t}).$$

5: Run Algorithm 3 on the conditional Feynman–Kac model with reference path $\bar{z}_{1:T}$, mutation kernels $\{M'_t(\cdot; \phi_t)\}_{t=1}^T$, and potentials $\{G'_t(\cdot; \phi_t)\}_{t=1}^T$, and return the resulting trajectory $\tilde{z}_{1:T}$.

As in the other auxiliary constructions, we first sample $\phi_{1:T}$ from the retained reference path and then run CSMC on the conditional Feynman–Kac model $\pi'_T(z_{1:T} | \phi_{1:T})$. The key point is that the conditional law of u_t given ϕ_t^u under the joint measure $\mathbb{W}_t(du_t) \mathbb{P}_t(d\phi_t^u | u_t)$ is again the pCN

kernel,

$$\mathbb{P}_t(du_t \mid \phi_t^u) = \frac{\mathbb{P}_t(d\phi_t^u \mid u_t) \mathbb{W}_t(du_t)}{\mathbb{W}_t(d\phi_t^u)}. \quad (2.9)$$

This follows from reversibility of $\mathbb{P}_t$ with respect to $\mathbb{W}_t$. Hence, conditional on $\phi_t = (\phi_t^u, \phi_t^e)$, the modified mutation kernel is

$$M'_t(z_{t-1}, dz_t; \phi_t) = \mathbb{P}_t(du_t \mid \phi_t^u) \mathcal{N}\left(e_t; \phi_t^e, \frac{\delta_t}{2} I\right) de_t. \quad (2.10)$$

The corresponding potential is chosen so that $M'_t G'_t$ recovers the correct augmented incremental target. Therefore,

$$G'_t(z_{t-1}, z_t; \phi_t) = Q_t^{\leftarrow}(z_{t-1:t}) \frac{\mathbb{P}_t(d\phi_t^u \mid u_t) \mathcal{N}(\phi_t^e; e_t, \frac{\delta_t}{2} I)}{\mathbb{P}_t(du_t \mid \phi_t^u) \mathcal{N}(e_t; \phi_t^e, \frac{\delta_t}{2} I)} \quad (2.11)$$

$$= Q_t^{\leftarrow}(z_{t-1:t}), \quad (2.12)$$

since the Gaussian term cancels by symmetry and the pCN term cancels by reversibility with respect to $\mathbb{W}_t$.

Thus, CD-pCN is therefore a standard CSMC update on an augmented state space, with mutation kernel obtained by combining a conditional pCN move for the Wiener component and a Gaussian random-walk move for the end-point component, while the potential remains exactly the original backward incremental weight $Q_t^{\leftarrow}$.

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